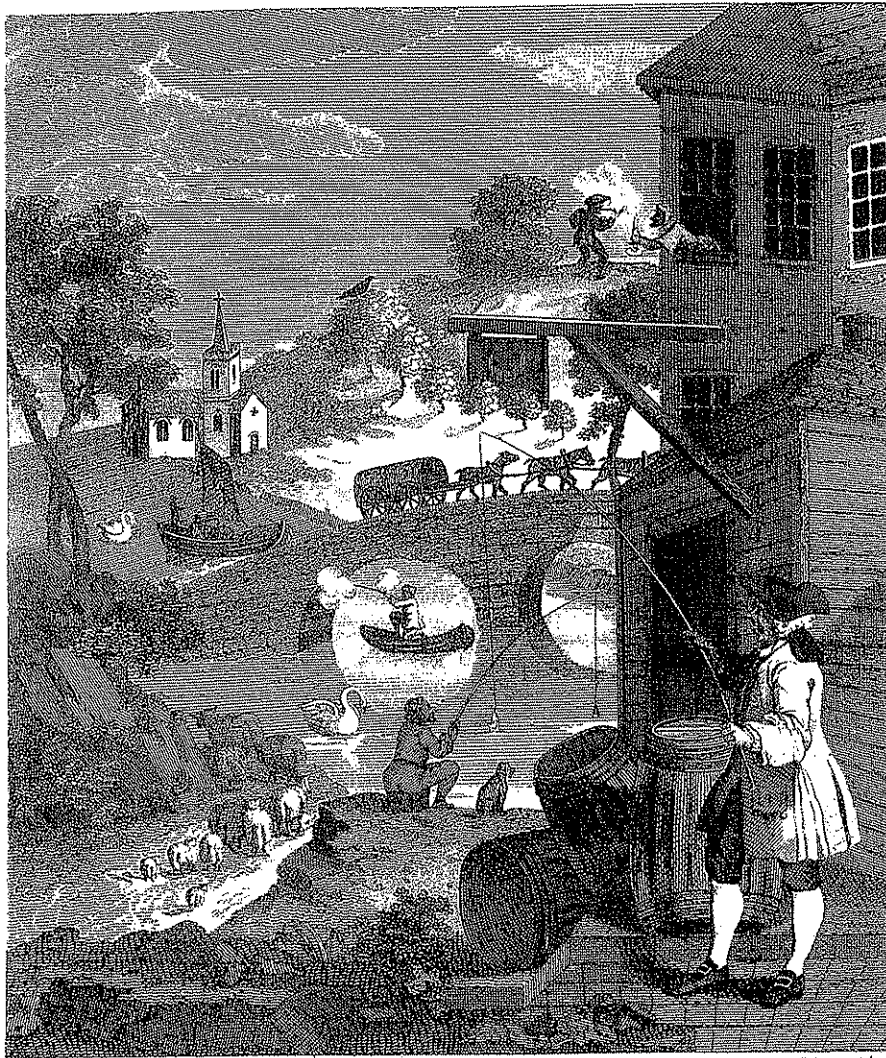


Critical Thinking About Research



Whoever makes a design without the knowledge of perspective will be liable to such absurdities as are shown in this frontispiece.

Critical Thinking About Research

PSYCHOLOGY AND RELATED FIELDS

Julian Meltzoff

American Psychological Association • Washington DC

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To Antonia

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Preface

This book was stimulated by my having spent the last 17 years teaching graduate students how to design and critique research. During this period I had the responsibility of critically evaluating far more dissertation proposals than I care to admit for fear of being too closely associated with Baron Munchausen.

Although there are many good texts on research design, they are written from the perspective of the research producer, not the research consumer. My colleagues and I have had to become inventive and innovative when teaching the important skill of critiquing research. From a pedagogical point of view, the need for supplying flawed research articles on which students could practice their critiquing skills was always apparent to me. The idea of creating the materials to order came later. Construction of such reading matter was informed by my experience in discovering errors that I repeatedly encountered when reviewing published articles as well as research proposals. I am indebted to all of the students and professional psychologists who unknowingly supplied me with the ideas that went into this book and who so graciously recognized and corrected the problems once they were revealed to them. My thinking about designing and critiquing research was enriched by frequent detailed exchanges of ideas with my academic colleagues. Their cogent observations, and sometimes their alternative ways of looking at things, stimulated me continuously to reexamine research-related issues, to fine-tune my views, and to reappraise my judgments.

I am most grateful to all of those who reviewed parts of this manuscript and offered helpful suggestions to improve its scope, its focus, and its clarity. I am particularly indebted to Mark Sherman, with whom I discussed research issues so frequently over the years. He helped shape my ideas about the subject and the ways it can be taught. He not only read the entire manuscript, but gave me detailed and expert feedback about it. I am truly grateful to Andrew Meltzoff for his frank, careful, and incisive critique and his ongoing interest and support in this effort. Without the constant encouragement and support of Antonia Meltzoff, I probably would not have been so inclined to begin this project or to have become so engrossed in carrying it through to completion. As each section was converted from thought processor to word processor, it turned into a prolonged show-and-tell in which she participated with a gusto that fueled my own esprit. Nancy Meltzoff was a most helpful resource when I dared to venture into her area of specialized knowledge. If, with so many who share my surname involved, writing this book is beginning to sound like a cottage industry, it is only because all of them are PhDs who are highly accomplished scholars, scientists, writers, and educators in their own right, whose judgment I trust, whose candor I can depend on, and whose input I value.

I have Peggy Schlegel, acquisitions editor at APA Books, to thank for the guidance and encouragement that she provided from the beginning. My special gratitude goes to Andrea Phillippi, development editor, for her careful reading and most helpful suggestions. I am deeply grateful to Ida Audeh, whose formidable editorial skills have done so much to polish the text. I very much appreciate the work done by all of the reviewers and editors of the manuscript in its incipient and later stages.

Introduction

How should scientific evidence be evaluated? After reading a research article, one person may be fully convinced by the evidence that is offered; a second person may think that the same evidence might be true but regards it, on balance, to be equivocal and feels obliged to suspend judgment; and a third person may disdainfully dismiss it with scatological words of unmistakable meaning. Their different judgments are informed by different views about what proof is, are influenced by their different beliefs about the evidence that is needed to establish proof, and are limited by the amount of knowledge that they have about the methods used to gather evidence in the search for truth. They are also affected by variation in their ability to think critically. I use the phrase *ability to think critically* as a positive quality, not as a pejorative reference to the characteristics of a mean-spirited faultfinder. The phrase refers to the skill that one has in thinking about an issue, analyzing it, looking at it from all sides, and weighing whether there is sufficient evidence of good-enough quality to warrant making a reasoned judgment that is as free of personal bias as possible.

As desirable as it would be to teach people to think critically, I am afraid that the best that I can do in a text of this sort is (a) to exhort people to adopt a critical mental set when reading research literature, (b) to try to increase their understanding of the principles and methods of research that are used to produce evidence, (c) to alert them to booby traps that can compromise the research and cloud the evidence, and (d) to provide them with materials that

can be used for practice, with the goal of improving and honing their critical reading skills.

This two-part book is about how to evaluate scientific evidence. Part I provides an overview of the methods and principles that one needs to know about when critiquing research. It is meant as an adjunct to (rather than as a substitute for) any of the excellent research methods books that are available at all levels. It is certainly not a statistics text, and it assumes that readers have some familiarity with basic statistical methods. It is not intended to prepare people to design and to carry out research. The focus is on reading and critiquing other people's research. The two activities are in no way contradictory, but a person who is adept at critiquing research is not necessarily skilled at thinking up research questions or doing research any more than a drama critic is necessarily a good playwright. Books on research methods aim to give people who are preparing research an insider's view of how to go about designing research and planning and conducting statistical analyses of data.

This is not a methods book for aspiring and practicing researchers. It is intended to assist the individual who reads and assesses published research. A reader who may not be familiar with all of the intricacies of research design (and who may not be aware of all of the things to avoid) should know enough to recognize procedures that have gone awry or to know when plans have not worked out optimally. Knowledge about the assorted "dogs that can come and bite" a researcher can help to guide the reader in a search for problems. I review the application of research principles and alert readers to many of the things that interfere with valid research. The objective is to help people bring inquiring and informed minds to the reading table. My presentation of the ways an experiment can founder, and the emphasis I place on scrutinizing and critiquing research, is in no way intended as an indictment of the research process or as an expression of any reservations whatsoever about its value as a method of seeking the truth.

I hope to encourage a form of interactive reading that I believe is central to the process of the critical evaluation of research literature, as opposed to the almost reverential acceptance of the printed word. The reader must be able to bring a skeptical, inquiring, searching, probing attitude to the task; to spot things that do not look quite right; and to recognize how a weakness in one phase of the design or execution of the study can severely compromise it at a later phase. It is analogous to conducting a comprehensive physical or psychological examination, but without benefit of any standard tests. The "tests" take the form of questions that the reader asks of the written research report.

Psychologists may spend their time in one of three ways: (a) they can do research to generate knowledge, (b) they can transmit knowledge to others by teaching or by directing the research of others, or (c) they can apply psychological knowledge in the form of clinical or consulting services. These can be done singly or in any combination. A parallel set of options is available to specialists in other behavioral or social sciences, in education, and in medicine and the health sciences. There is variation between and within fields in how much time is spent educating and training students for the functions of generating knowledge, disseminating it, or applying it. One thing shared by people in all fields and subspecialties, no matter what career direction they choose, is that they all have to be consumers of research. The bedrock of psychology and other related fields is made up of theories and data from research that support the theories. All who profess to know a field, and those who want to keep abreast of advances, are necessarily consumers of research. It stands to reason that they must be able to evaluate research critically. Having a good grasp of the scientific method and the principles of research design and knowing what to look for are invaluable in this pursuit.

Psychologists who graduate from research-oriented doctoral programs have the advantage in this regard because research knowledge is a vital part of their education, and the dissertation requirement gives them hands-on experience in research. Nevertheless, virtually all programs that do not require a research dissertation strive to teach students to think critically and to read empirical work intelligently. Nobody who is considered an expert or specialist likes to be the bearer of misinformation. Throughout the history of psychology and medicine, untested fads, half-baked ideas, and outright quackery have blown across the land, sweeping the ill informed along and leaving human wreckage in their path. Many professionals with weak critical-evaluating skills have become enthusiastic and authoritative advocates of unsubstantiated procedures. Other professionals have had the wherewithal to resist and have retained their skeptical reserve until solid research data were furnished.

What is necessary, then, is first to learn how to critique research and then to make a habit of sifting all incoming offerings through a sieve whose openings are small enough to permit only the truths or near-truths to fall through. This book is written to further that goal. I present the principles of research design that people need to know about in order to critique research (part I), followed by intentionally flawed, but realistic, fictitious materials on which to practice critiquing skills (part II).

Chapter I deals with the methods of seeking the truth and with the rules of evidence and what constitutes proof. It discusses how to

read critically and presents an outline to serve as a guide for critiquing research. Chapter 2 focuses on the research question and hypotheses and discusses faulty reasoning about causation. Chapter 3 reviews the critical appraisal of research strategies and the selection of variables. The selection and composition of the research sample, methods of sampling, and the consequences of faulty sampling are covered in chapter 4. Chapter 5 gets to the heart of research design in a discussion of the variables that should be controlled and methods of controlling them. Chapter 6 presents the major experimental and quasi-experimental designs that are available and discusses the flaws that the reader should recognize in some of them. It also gives examples of some of the important threats to the internal validity of a study. The use and misuse of criteria and criteria measures are dealt with (chapter 7) to facilitate the reader's appraisal of how these techniques were selected and used in a particular research study. Chapter 8 is an orientation to the assessment of statistical analyses, graphic analysis, and the drawing of inferences and conclusions. Ethical considerations in research, the subject of chapter 9, are discussed to alert the reader to deviations from ethical standards. All of these chapters in part I are written from the perspective of the reader looking in on the research report from the outside, with the emphasis on critically evaluating research rather than doing it.

Not covered are some techniques, such as meta-analysis, that are important for critical readers to know about if they are trying to pull together a connected series of studies. Some students know how to critique individual studies but have difficulty integrating a series of studies and weighing the balance. Although this is an important skill, it is beyond the scope of this book. Appraisal of the individual study comes first. There is not much point including a badly flawed study and giving it weight in an overview; it should just be dismissed.

Part II consists of 16 original, realistic, but entirely fictitious simulations of journal articles. The articles have flaws built into them that range from the blatant to the subtle. The collection is essentially a workbook that is intended to be used by readers to practice critiquing skills that are based on the principles in part I. The frontispiece by the great 18th-century artist, Hogarth, was one of the things that gave me the idea to write this book. The legend at the bottom of the print says, "Whoever makes a design without the knowledge of perspective will be liable to such absurdities as are shown in this frontispiece." So it is with research. Lack of understanding of the principles of research design can lead a researcher to the kinds of errors that are illustrated in the fictitious articles and can prevent a reader from realizing that there is anything wrong.

The articles in part II range across a variety of content areas (clinical psychology, general psychology, psychology and the law, education, and sociology) and research methods (experimental, quasi-experimental and single-case designs, and qualitative and survey research), but the predominant emphasis is on psychological experiments. Each fictitious article is accompanied by a critique of its built-in flaws. Critical issues are cross-referenced to the chapter in part I that addresses them, so that readers who want a fuller explanation will know where they can find it.

The kinds of statistical analyses in the articles range from simple descriptive statistics to univariate analyses of variance with categorical levels. Studies that use the techniques listed in the fictitious articles are abundant, and the articles mirror those that can be found in the literature. The articles do not feature highly advanced statistical techniques so that the book will be useful for the greatest number of people.

These critiquing exercises are meant primarily for the use of students in psychology and related fields, whether they be doctoral or master's degree candidates or advanced undergraduates. Psychologists whose graduate school experience is well behind them, and who have not themselves been engaged in research, can profit from a refresher. This is especially so for those who find that their skills have eroded from disuse or those who are overburdened by the need to keep abreast of a constantly expanding literature.

Because the principles of research design and the approaches to critiquing research are not restricted to psychologists or psychological research, students and professionals in other behavioral and social sciences may find these materials and exercises useful. This book also may be helpful to medical researchers, consumers of medical research, and members of the legal profession who deal increasingly with the evaluation of evidence and opinions that are offered by scientific expert witnesses. With so many scientific specialties, cross talk among them is necessary if one is to function intelligently. A person's understanding of research in another field may be circumscribed by lack of technical knowledge, but one should still be able to read and understand research in a related field.

In this age of science, the media continuously expose nonscientists to science reports from all fields. As Scarr (1996) has recommended, "we need to turn policy makers and the citizenry into information-consuming, psychological scientists" (p. 2). The ordinary person is served a rich diet of polyscience every day. The portions, unfortunately, take the form of sound bites that have to be swallowed whole or summaries that lack all of the details. Educated nonscientists who want to learn more about research design and how to read

research critically could profit from selected reading in part I, skipping sections that appear to be too technical and specialized. As information consumers, they could derive benefit from testing their critical reading skills on the exercises in part II.

Readers who are graduate or advanced undergraduate students can assess their own level and needs. Those who have already had considerable training in research design should select any sections of part I that will fill the gaps. Readers should evaluate the articles in part II on their own before reading the accompanying critiques. Concepts that are relevant to the article are cross-referenced with part I; readers should consult part I for further clarification.

Professional psychologists who have not done any research or who have done some research in the past but have been away from it for a long while may choose to look through the contents of part I and to start to read wherever they see a need for a refresher or whatever strikes their interest. Those who evaluate the articles in part II independently and find that they have missed many of the central points in the critiques may benefit from reading part I more thoroughly.

Readers who believe that they already have a good understanding of research questions, hypotheses, and sampling but that they could use a booster on design options and methods of control might want to read chapters 5 and 6. Those who are comfortable with various aspects of design but have questions about criteria and criteria measures could start with chapter 7. In either case, readers should try to sharpen their reading skills by critiquing the articles in part II.

Knowledgeable and experienced researchers who already know far more than is covered in part I may begin with part II and read the fictitious articles for fun, for challenge, and for mental calisthenics.

PRINCIPLES

Critical Reading

1

Critiquing research is not something that comes naturally. People (including psychologists) spend many formative years learning and using other ways of appraising truths. They are governed by other ways of evaluating the validity of people's claims before they learn about the scientific method and before they gain an understanding of the methods that are used in psychological research. First learned, deeply ingrained modes of thought continue to affect people as they go about the task of evaluating what they are told orally and in print and to influence them in deciding what evidence to consider valid.

This chapter reviews the various methods that people use for seeking the truth and their differing views on what constitutes proof. These views naturally affect the reader's attitudes toward the material under review and determine whether the material is accepted uncritically or is subjected to critical appraisal. The crux of the critical appraisal, in turn, depends on the rules of evidence that the reader holds necessary and the reader's knowledge of research design.

Methods of Seeking Truth— Proof and the Rules of Evidence

Diverse methods of seeking truth and varied codes of rules of evidence may conflict, coexist, or even work in harmony

with each other for the same individual in different spheres of life. What works for a person in one aspect of life may not be satisfactory in another. Some people cannot "change hats" easily; others can switch seamlessly from one mode of thought to another, as occurs, for example, when one reads a historical novel, interacts socially, goes to church, and reads scientific studies.

FAITH

Authoritative Pronouncements

Some people take the judgment of one or more authorities as sufficient evidence. This can take the form of ex cathedra pronouncements by religious figures or authoritative pronouncements by experts. Experts can be highly reputable scientists whose credentials give cachet and weight to their pronouncements. What if two or more experts disagree, as they so often do? With the unerring benefit of hindsight, one can see how mistaken highly qualified scientific experts have been in their pronouncements. In *The Experts Speak: The Definitive Compendium of Authoritative Misinformation*, Cerf and Navasky (1984) provided retrospectively amusing quotes from outstanding authorities in various fields. The physicist Lord Kelvin, who was president of the British Royal Society, announced in 1895 that "heavier-than-air flying machines are impossible" (p. 236). In 1897 he stated that "radio has no future" (p. 206) and in 1900 claimed that "X-rays are a hoax" (p. 302). In 1878 Professor Erasmus Wilson of Oxford said, "[W]hen the Paris Exhibition closes, electric light will close with it and no more will be heard of it" (p. 203). Einstein is reported as having said in 1932 that "There is not the slightest indication that [nuclear] energy will ever be obtainable. That would mean that the atom would have to be shattered at will" (p. 215). In 1956, the leading British astronomer Richard van der Riet Wooley allegedly said, "Space travel is utter bilge" (p. 258).

Democratic Judgments

Placement of faith in the judgment of many in contrast to the dicta of single authorities is a common way of deciding what is right. The saying goes, "Fifty million Frenchmen can't be wrong." Confidence is instilled when a judgment is shared by many people. Committees decide what is true (and the blame can be shared among them when they are proven wrong).

REASON

Another way of arriving at truth is by reason alone. Under certain circumstances pure reason can lead to what is called *necessary truth*. In a

priori reasoning, conclusions are deduced from known and self-evident principles or definitions, such as "all men are mortal." The syllogisms of formal logic and the proofs of pure mathematics lead to necessary truth.

However, not all questions can be answered by pure reason alone. If they could, there would be no need to design or carry out experiments. Empirical work calls for an inductive a posteriori process whereby final judgments are based on experience, and principles are generalized from facts that are established by experience rather than by the application of reason alone.

FEELING

Ad hominem arguments appeal to the feelings and passions rather than to reason and intellect. The person asserts subjective truth in saying, "I feel it to be true in my heart, in my guts. Therefore, it must be true." The determination of objective truth from gut feelings is problematic when two people have equally intense but opposite feelings.

SENSORY INFORMATION AND EXPERIENCES

One of the most compelling and unshakable proofs for people is personal experience. There is a certainty to the subjective truth of the statement, "I have seen it, I have experienced it. Therefore, I know that it is true." The reality of extraterrestrial presences is firmly established for people who believe that they have witnessed, sensed, or experienced them. One step removed from such proof is the acceptance of the testimony of others who claim to have sensed or experienced something. Scientists, who are trained to be skeptical, are not immune to the importunity of personal experience. The famous self-deception of Professor René Blondlot (1905), the noted French physicist, who thought he saw spectroscopic evidence of nonexistent "N-rays" being emitted from metals and who convinced the scientific community of their existence, is an instance of this. The fallibility of the human senses is well-known to psychologists who have studied illusions and to magicians who exploit this fallibility to their advantage.

LEGAL METHODS

Legal methods for arriving at truth call on all of the above methods including authoritative testimony of experts, appeals to pure reason, ad hominem arguments, and testimony of witnesses about what they have seen, heard, or experienced. Faith is then placed in the judgment of the jury members who weigh the evidence and decide the truth. A

simple democratic majority is not sufficient for conviction in criminal cases that demand unanimous agreement.

EMPIRICAL AND EXPERIMENTAL METHODS

Empirical (which include experimental) judgments are based on objective and systematic observations or experiments. In experiments, the scientist exercises control over the variables by careful prearrangement of conditions. In nonexperimental empirical work, the scientist observes and records but does not manipulate the variables. These approaches to discovering or establishing truth reflect a basic skepticism about proof and are characterized by a "show me," "prove it," and "what is the evidence?" attitude about alleged "facts." Although the conclusions that one reaches are neither subjectively nor objectively certain, one can judge how probable they are. Even though absolute certainty may be beyond reach, one does aspire to attach a level of probabilistic confidence to judgments about what is true.

Approach to Reading Critically

REVERENCE FOR THE PRINTED WORD

In the days when few people could read, written symbols or words were painstakingly carved on stones, walls, pillars, and clay tablets. Anything that had to go through such a laborious process and was obviously being recorded for eternity must have seemed to viewers to be of utmost importance. Publication in this mode did not invite critical scrutiny, nor did it serve as a prelude to dialogue.

Publication requires either a one-of-a-kind public display or the dissemination of multiple copies. There was an advantage to the use of soft and flexible materials, which could be posted, copied, transported over distances, or stored in libraries. The printing press made it far easier to make alterations. Words could be added or deleted and sentences or paragraphs moved about. Paper itself became less durable, as if to acknowledge that the written word is not permanent, that ideas do change, and that new truths come forward and old ones have to be revised or replaced. Obsolete documents, as distinct from stone, can easily be crumbled or torn up and discarded.

As literacy became more common, and the means of disseminating thoughts through the written word became available to more people, previously entrenched truths were increasingly challenged. People not only could read, but they could privately disagree with, and even

publicly rebut, what they read if the social climate of the time was sufficiently enlightened to permit it. New writing instruments such as lead pencils and quill pens made it possible for ordinary people to write down their thoughts. The eraser, conveniently placed at the end of the pencil, accommodated an expectation that errors of thought and inadequacies of expression would occur. Things written in pencil have a tentative and impermanent air about them. The fountain pen, an invention of the 19th century (Ansley, 1935), liberated people to leave their more indelible mark anywhere. During its brief history, the typewriter made it possible for anyone to produce official-looking printed words and to make rapid changes if desired. Now that computers have become the writer's choice, whole sections can be deleted with the press of a button or moved about thanks to the miracle of the cut and paste function keys.

From this brief and sketchy sweep through the history of writing, a few things stand out. Worthwhile thoughts have not gotten any easier to come by, but it is certainly simpler to record them and to communicate them to others. When almost anyone can print out words and thoughts, there is, I suspect, a decline in reverence for the printed word. This sets the stage for critical reading and makes it a more natural and acceptable thing to do.

CRITICAL READING

The critique of what is presented as a scientific study can justifiably address the way the study does or does not meet the scientific standards for evidence and proof. In reading such a study, the reader's focus is on its scientific soundness, not on whether the findings conflict with preexisting faith, beliefs, or ideas about social acceptability; not on whether the results conflict with expert opinion or clash with other methods of establishing truth and gathering evidence; and not on whether the results are in harmony with his or her gut feelings. The critique mirrors the reading itself.

Even though it cannot provide more than probable truth and must be taken with caution, empirical (and particularly experimental) research is still the best way to look for answers to certain types of questions. Presentation of all of the ways an experiment can founder and the emphasis that this book places on scrutinizing and critiquing research is in no way intended as an indictment of the research process or as an expression of reservation about its value as a method of seeking the truth.

Doing research and reading research reports critically are both arduous processes. It is much easier to say, "This is too hard, let's just brainstorm it and take a vote," or "Let's just ask Dr. Jones, he ought

to know," or "Let's just review the literature and take what the experts say," or "Let's hold a symposium, invite some experts to debate the question, and give the nod to the one who is most persuasive." People may become so overwhelmed with all of the difficulties associated with doing and critiquing research that they detach themselves from it and declare that they are simply not interested in this method of finding things out or that they find it too problematic and too inconclusive. Others admit that it is too difficult and too technical for them to judge by themselves and that they prefer to leave it up to the researcher, who, after all, must know best. They then take, on faith, whatever is printed in the scientific journal. This defeats the whole purpose of presenting research for public scrutiny.

The phrase "research shows" is as revered in some circles as ex cathedra pronouncements are in others. The important thing is for the person who utters these words to have read the research, to have understood it, and to have evaluated it before using its findings to support a position or to inform a practice. When the research is truly sound, however, there is no more powerful way to make a point than to cite research findings.

Critical reading requires a mental set of a particular kind. This mental set can be taught, encouraged, and nurtured. Conversely, it can be discouraged or even forbidden. What is involved, first of all, is a kind of general open skepticism that enables one to bring a "show me" or "convince me" attitude to the reading table regardless of how authoritative the author may be or how attractively the words are packaged. The reader then engages in an interactive dialogue with the manuscript. The critical reader applies exceptional focus to each sentence that is written, contemplates the meaning behind it, and thinks about the thoughts that were not written and wonders why they are absent. This cannot be done in haste, and it often requires multiple rereading and rumination about particularly troublesome sections. Critical reading of scientific materials is very different from recreational reading of science fiction or of a novel (although these too can be read critically by literary scholars). When reading scientific documents, every thought that is expressed, every conclusion that is reached, has to be screened through one's internal skeptical scanner. If everything meets the most stringent cognitive challenges, one comes away enriched and gratified. On the other hand, one may accept some of it but have reservations about other parts. A third scenario is that one may reject it in its entirety and resent having spent the time on it. None of these scenarios is the same as reading in a passive-receptive mode, with one's critical antennae down, and accepting the author's conclusions wholesale.

Uncritical acceptance of conclusions leads to the incorporation of misinformation into one's body of knowledge. It is no trivial matter when such material misinforms public policy or becomes a false guide to one's clinical practice. "Research shows" is one of the favorite expressions of psychologists who are called on by the media to express their professional opinions on a wide range of topics, who are asked to consult with or testify before lawmakers about social issues that affect public welfare, or who are relied on to give expert counsel to other health service providers or to educators. Research psychologists thus carry a heavy burden of responsibility for assuring the accuracy of their claims about their results. In turn, psychologists who cite or apply the research findings of others share that responsibility. They have an obligation to use their critical reading and evaluation skills in reviewing a study before they cite it as evidence that supports a point of view and before they apply the findings in their clinical work. Uncritical acceptance of invalid research can impede the development of the field and jeopardize human welfare.

How different this process of critical reading is from gaining scientific information from summaries, abstracts, or digests, as busy professionals are often tempted to do! These provide conclusions to be accepted on faith alone. Yet nobody has the time to read all of the scientific and professional journals in the field. Digests and abstracts do serve a useful purpose. The best way for readers to use them is as a screening device for subject matter that may be of special interest, to be followed by critical reading of the actual articles. When faced with an original full-length article, experienced critical readers often skim through the article quickly so that they can get a general overview of the contents and can decide whether they are sufficiently interested to read it more carefully. If they decide to engage, they then begin to reread the article in a fully focused critique mode. Sometimes several readings of crucial sections are required. This is particularly true when the text is fuzzy or lacks readily accessible detail.

Interactive Reading

The passive reader is a recipient who takes in what the text says. At the end, the reader may have a general impression of the research and some points of agreement or contention. The interactive reader anticipates what is to come and then discovers whether these expectancies are met along the way. Adoption of the interactive mode places the reader in the best position to evaluate the work critically. Throughout this book, I call attention to the kinds of expectancies that flow naturally from given antecedents:

- that the research question guides the literature review
- that the literature review and the statement of the problem inform the hypotheses
- that the hypotheses set up design expectancies and suggest what variables should be controlled
- that understanding what should be controlled influences the design and the procedures and prompts a reader to look at the methods that are used to exercise control
- that the hypotheses and design and type of data dictate the method of data analysis
- that the analysis of the data influences the kinds of conclusions, inferences, and generalizations that can be made.

The interactive reader is working all of the way, checking evolving expectancies against accomplishments in the study. The critique, whether negative or full of praise, can be focused and justified.

When in the interactive reading mode, the reader selects an article to read because the title sounds as though it might be of interest. Seeing the title, "Effect of Teaching Method on Reading Ability," one expects to find a cause-effect experiment in which the method of teaching is systematically varied, and reading ability is compared in groups of children taught by different methods. The review of the literature and the statement of the problem reinforce that expectation, and the hypotheses are worded in a causal manner. Turning to the Method section, the reader is soon disappointed to find that instead of manipulating teaching method systematically, the researcher has located different elementary schools within a large school system. The method of teaching reading varies from school to school. One immediately wonders about the equivalence of the various groups of students to start with and the equivalence of the skills of the teachers. This sets up yet another unfulfilled expectation that the researcher has made sure that the groups are equivalent and that the teachers are comparable. Proceeding, the reader anticipates a valid and reliable measure of the dependent variable, reading ability. Instead, supervisory rankings are made of the various classes, and testimonials are given about the accuracy of this ranking procedure. Turning to the analysis of the data, the reader expects, in view of the ordinal data, that nonparametric statistics are used. Again, expectancies are not met; the author uses parametric statistics instead. Curious about how all these issues are handled in the Discussion section, the reader finds unjustified authoritative pronouncements and unfounded causal statements in support of the hypotheses.

How different is this interactive reading process from the passive-receptive mode where the reader sees the article as excessively

detailed and would much prefer to read the abstract and skip the rest. Good "critical" interactive reading does not require the uncovering of crucial flaws. A special pleasure and admiration come to the interactive reader when serial expectations are met, and in fact exceeded, when the researcher anticipates and forestalls problems that the reader has not thought about, when the research procedures not only fit basic requirements but do so in a particularly clear and original way, where the data are analyzed in an especially convincing manner, where the inferences from the data are unquestionably sound, and where the discussion is impressively profound and thought provoking.

At one extreme, the critical reader of sound scientific material will say, "I've learned something and will incorporate it into my knowledge base, I'll accept the conclusions and will cite them and apply them when appropriate." At the other extreme, the reader of a flawed study cannot accept any of it as true.

Knowledge of Research Design

Readers may sit down in a state of heightened alertness, eager to critique an article of interest. Are they prepared? Do they know what to look for and on what to focus their attention? Will they see the defects or be oblivious to them? If readers do not have the background and training to critique the article intelligently, they are bound to have misgivings. They are in a position analogous to that of a person who is contemplating the purchase of a used car. Should faith be placed in the salesperson's appraisal of the car's merits? Does the buyer know how to assess it? Does he or she know all of the potential problems that may be present in the car's mechanical and electrical systems and how to evaluate their presence or absence? People who lack the specialized knowledge can always hire an expert to advise them. Many people rely on drama and motion picture reviewers to screen the offerings and to offer suggestions about what is worthwhile. When it comes to scientific journal articles, however, there are usually no reviews to consult. Buoyed by the knowledge that articles in some journals have been previewed by referees and read by an editor, readers may feel confident that the articles have been pasteurized and are safe for consumption. Nonrefereed journals do not even offer the added protection of a preview.

This kind of prepublication review in select journals is a helpful form of quality control. In 1994, the rejection rate of articles submitted for publication in American Psychological Association (1995) journals averaged 75% and ranged from 44% to 92%. Regardless of how stringent this appears, no article that succeeds in being published is accompanied by a guarantee of excellence. How many times has one

seen flawed articles in refereed journals? How many times has one read critiques of an article, followed by the author's rejoinder? Is it not commonplace, and in fact expected, that authors of journal articles critique the work of their predecessors? This would never be necessary if all articles in refereed journals were beyond criticism. Obviously this is not the case.

Any research article has three components: (a) the technical content as reflected in the literature review, statement of the problem, and formulation of hypotheses; (b) the research itself; and (c) the interpretation and discussion of the results. A reader who knows much about a topic, but little about research design principles, is not in a good position to evaluate a research article critically. A reader who knows little about the topic can nonetheless evaluate the design and execution of the research. Such a reader may miss out on some potentially crucial details regarding the choice and use of specialized apparatus or the selection of the best available criterion measures or most advanced techniques. Generally, however, readers who review material that is within their broad general sphere of knowledge should be able to critique any but the most highly specialized articles.

Principles of research design transcend content areas. The effectiveness of a reader is therefore dependent on knowledge of research design and on skillful application of that knowledge. The remaining chapters in part I discuss those aspects of research design that are essential to know about if one is to conduct a comprehensive appraisal of a research article. A systematic approach to the critique of research reports is outlined in the Prologue to part II (pp. 161–167).

Research Questions and Hypotheses

2

The Research Question

In this chapter, I examine the principal types of research questions, give examples of each, and review the design expectations that each type of question establishes. I then address the complex and controversial matter of causation and raise issues that must be resolved whenever the research question is of a causal nature. Finally, I discuss the importance of providing a definition of terms and formulating clear hypotheses.

Research can be regarded as a process of asking a question (or a related series of questions) and then initiating a systematic process to obtain valid answers to that question. In reporting the research, the question should be made clear to the reader. If it is not explicitly stated, the reader ought to be able to infer what the question is from reading the introductory material in the text. It is disconcerting to read research reports that reflect the author's apparent confusion about the precise question or questions that the research is meant to address. This sometimes leads to hypotheses that do not seem to be consistent with the question, is followed by procedures that do not test the hypotheses, and culminates in conclusions that may not be too closely connected to the original question. The question can be directly posed, or it can be incorporated in a statement of the problem. The statement of the problem

